

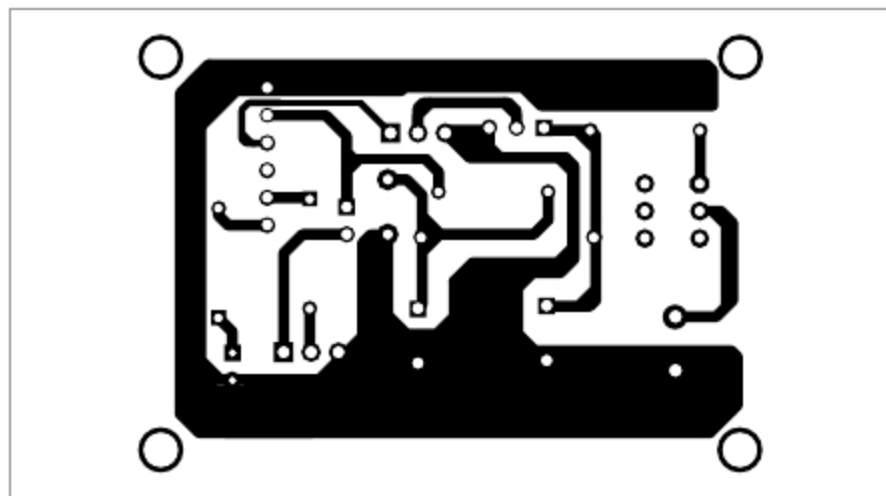


Fig. 3: Actual-size PCB of motion detector security alarm

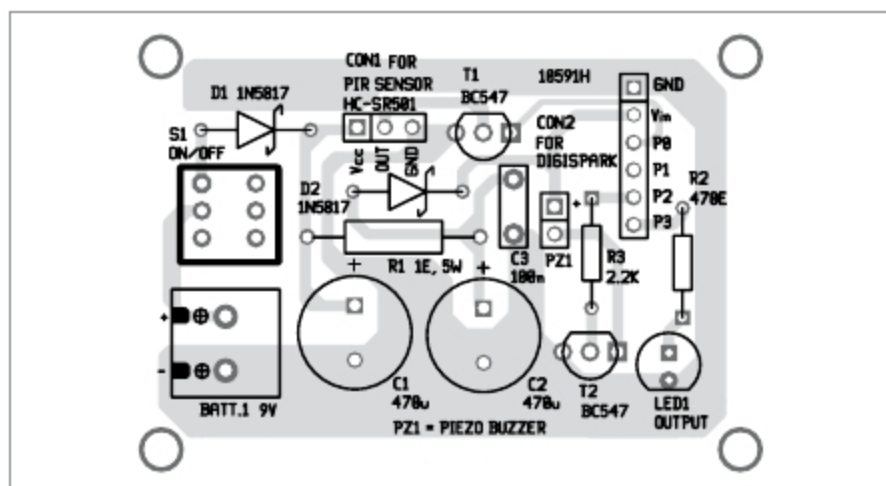


Fig. 4: Components layout for the PCB

Fig. 3 and its components layout in Fig. 4.

Construction can begin by first mounting the discrete components on the PCB or perfboard, followed by the motion sensor and Digispark module. Note that, connector CON1 is for the PIR sensor, and CON2 is for Digispark.

You can experiment with onboard potentiometers in the motion sensor module (HC-SR501) to adjust the timing period (delay) and detection threshold (sensitivity). You can also change the delay in the program (digi_motion.ino).

After construction, enclose the circuit in a suitable box and fit the power switch on the back of the enclosure, as shown in the prototype (Fig. 1). Connect a 9V battery for power supply to the circuit.

After selecting a location for the installation of the circuit, turn on the switch (S1) and wait for 60 seconds for the PIR sensor to initialise. LED1 will flash intermittently to indicate the system status. When the circuit detects motion, the piezo-buzzer (PZ1) will make a sound and LED1 will start blinking rapidly.

Keep the circuit away from heat sources such as radiators, heating ducts and stoves. Also, do not expose the PIR sensor to sunlight or rain. If the alarm sounds at random, it could be because the device is located too close to a heat source. In that case, change the location or direction of the sensor unit. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com



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